

Every Face Has a Story

Automated die studies at civilisational scale:
estimating the monetary output of the Roman Empire
from millions of unprovenanced coin images

1. Abstract

2. Introduction

3. Related Work

3.1. Die Studies in Numismatics

3.2. Deep Metric Learning for Cultural Heritage

4. Data

4.1. Training Data: Schaefer / RRD

Richard Schaefer’s Roman Republican Die Project (RRDP) provides our ground-truth labels: 77587 catalogued specimen-faces, spanning 16967 hand-identified dies across 445 RRC issues (7941 obverse, 9030 reverse). We have downloaded 70371 single-face crops (median 464×455 px; 10th-percentile short side 409 px) and trained the embedding on the bulk of this corpus.

The die-size distribution is the regime that governs output estimation: 34.5% of obverse dies are seen exactly once and 50.2% at most twice, while the largest die is represented by 72 coins. For benchmarking we use the 89 obverse issues with at least 50 coins and 15 dies (34175 coins). Near-duplicate imagery is handled by the dedup pass.

4.2. Deployment Data: sixbid archive

The deployment corpus is the sixbid auction archive (12.4M lots). A high-recall prefilter keeps 2.58M lots with any ancient/Roman signal; of these, 601528 lots cite a catalogue and were downloaded as full-resolution two-face photographs. Because no die study can proceed without knowing which coinage a coin belongs to, every lot is first assigned a **catalogue type**: a CRRO (RRC), OCRE (RIC) or RPC identifier. Citations in the lot description resolve 459924 lots to a type; those labels train an image model (shared DINOv2 ViT-B over obverse and reverse, cosine classifier, nearest-neighbour assignment against the labelled gallery) which reaches 0.94 / 0.823 / 0.646 held-out exact-type accuracy on RRC / RIC / RPC (top-5: 0.964 / 0.915 / 0.754), and refuses non-Roman coins (Greek, Byzantine, Celtic, world) at a false-annotation rate below 1%. Applied to the uncited remainder of the candidate pool it annotates a further 355949 lots, for 930759 typed coins in total: 175303 RRC (1543 types), 621147 RIC (20542 types) and 134309 RPC (13574 types). Figure 1 summarises the corpus. The type layer supplies the population each die study is run within, the dated denominator for every issue, and the structural match between the deployment pools and the Schaefer benchmark pools.

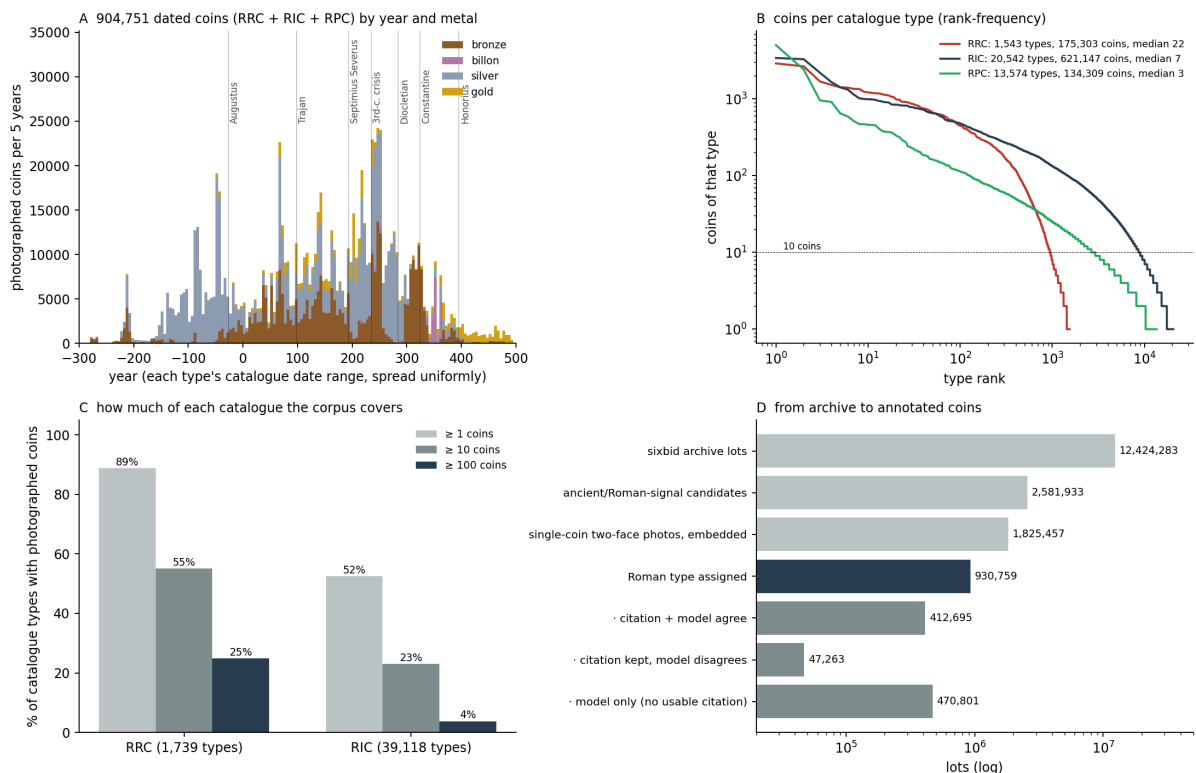


Figure 1: The annotated corpus. (A) Dated coins by 5-year bin and metal; each coin is spread uniformly over its type’s catalogue date range (interval data, no lot midpoints). RRC and RIC dates come from CRRO/OCRE; RPC Online publishes no bulk data, so its type records (province, city, reign, dating, metal) were collected page by page, with reign-level dates where a type has no explicit dating. (B) Coins per catalogue type: RRC is head-heavy (median 22 coins per type), RIC and RPC are long-tailed (medians 7 and 3). (C) Share of each catalogue’s types with at least 1, 10 and 100 photographed coins. (D) From archive lots to typed coins, by annotation source. “Coins” are auction lots; resales are not yet collapsed at this level.

5. Method

The pipeline has five stages, and the type layer is what makes it short. Because every coin carries a catalogue type, the corpus is a set of closed pools rather than one open universe, and the stages that exist to cope with an open universe — approximate nearest-neighbour search, clustering, coverage estimators — are not needed.

5.1. Segmentation and Embedding

Auction photographs show both faces of a coin side by side on a plain ground. A foreground mask splits them at the valley between the two discs and maps each face onto a square canvas, coin centred, padded with the background colour, so that a rim clipped on one side is recovered from the other (`gpu/coin_crop.py`). Faces are embedded twice: once for type, by a DINOv2 ViT-B/14 over both faces jointly, and once for die identity, by a DINOv3 ViT-L/16 with LoRA adapters at 518 px trained with ArcFace on Schaefer’s die labels. A single backbone with two heads reaches the same type accuracy but loses 6–8 points of die-gate recall at 224 px, so the two embeddings are kept.

5.2. Typing and Candidate Generation

The type model assigns a catalogue identifier and refuses non-Roman coins (§4.2). Within a type, every pair of coins is scored exactly by embedding cosine and every pair above 0.6 is passed to the verifier. A closed pool of a few hundred coins makes this cheap — 6693446 pairs for the Republican corpus — and it removes the candidate-selection step that an approximate index would otherwise impose, along with the recall term it would add to the model.

5.3. Verification

Two coins struck from one die share the relief exactly, so their keypoints align under a single rigid transform. DISK extracts up to 2,048 keypoints per face; LightGlue matches them; RANSAC fits a partial affine transform, and the score is $vr = \text{inliers}^2 / \text{matches}$, fused with the embedding cosine as $vr + 100 \cos$. The same alignment, read on the flan outline instead of the relief, identifies the individual coin and so detects resales (§4.3).

5.4. Counting Dies Without Clustering

Cluster counting fails in one specific way: a single false link merges two dies, and enough false links percolate into giant clusters, so single-linkage is only safe at an edge precision of 0.999. The quantity we want can instead be read off pairs. Within a type whose dies carry shares p_i , two coins drawn at random share a die with probability $q = \sum p_i^2$, and $D_{\text{eff}} = \frac{1}{q}$ is the effective number of dies, the same object as the effective population size in genetics. Estimating q needs only the fraction of within-type pairs the verifier accepts, corrected by its recall and false-positive rate measured on Schaefer (leave-one-type-out); a false link is then a linear bias, never a merge. Figure 2 shows the calibration and the result: on the 10 bench types with at least 50 coins, the pairwise estimate sits at 0.917 of the truth with cv 0.238, flat across thresholds, against 1.648 (cv 0.217) for single-linkage at the same threshold. The residual spread is the verifier’s per-type recall, which is the remaining lever.

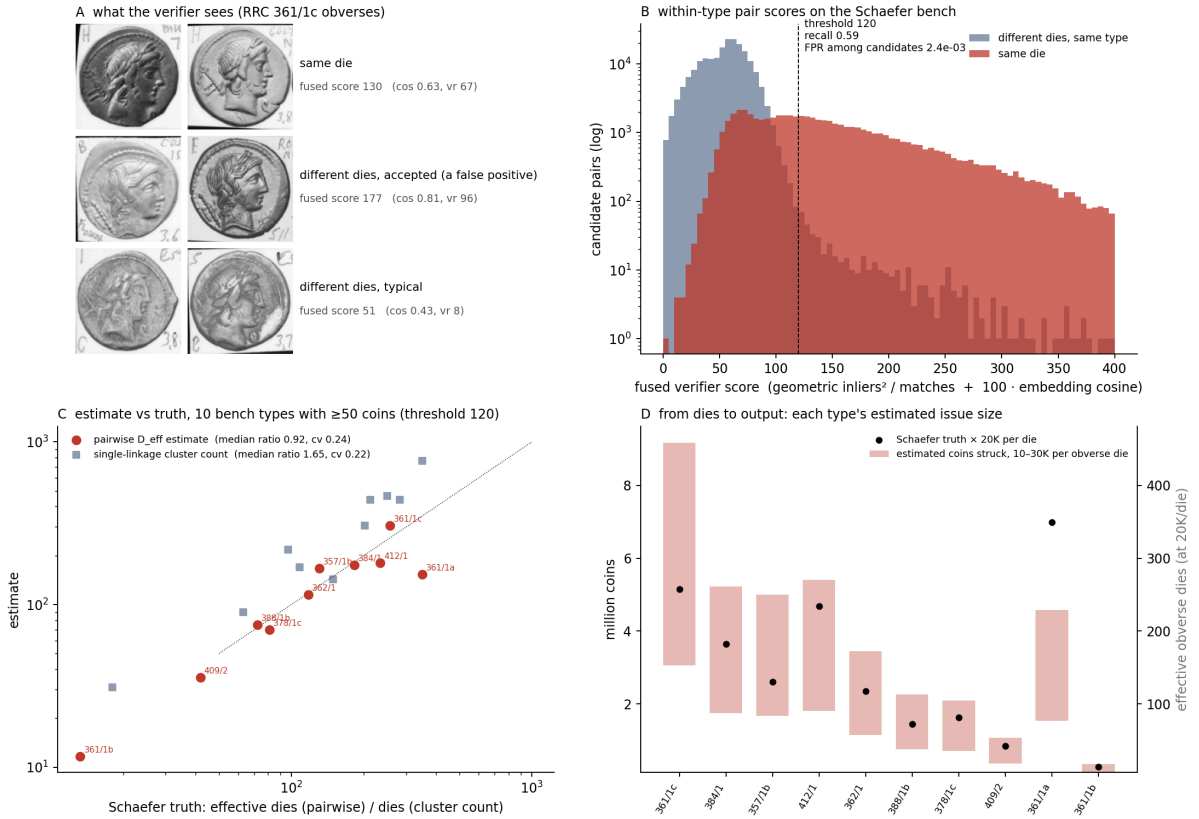


Figure 2: Counting dies without clustering. (A) Within one type, a same-die pair, a different-die pair the verifier accepts at the operating point, and a typical different-die pair, with fused scores. (B) Within-type pair scores on the Schaefer bench; at threshold 120 recall is 0.586 and 0.00244 of candidate pairs are false positives. (C) Pairwise D_{eff} estimate against Schaefer truth per type (leave-one-type-out calibration), with the single-linkage cluster count at the same threshold. (D) From dies to output: D_{eff} times the conventional 10000–30000 coins per obverse die, per type; the coins-per-die assumption dominates the uncertainty.

5.5. Republican Silver Output from Die Matches

Applying the pairwise estimator to the sixbid corpus gives the first die-based output series for the whole Republican denarius coinage, with every source of uncertainty propagated. Every same-type candidate pair among the 99050 RRC-typed lots was geometrically verified (6693446 pairs, both faces). Because a typed corpus is a set of closed pools, candidates need no approximate search: every pair inside a type with embedding cosine at least 0.6 is verified, and the same gate defines the bench.

An auction corpus resells coins, and a resold coin photographed twice is a perfect same-die pair that is not two coins. One mechanism separates the two cases: the verifier’s alignment is read twice. Relief geometry identifies the die; the flan outline identifies the coin, because a hand-struck flan is irregular and unique while two coins from one die have different flans (Figure 3). Provenance citations in descriptions (“Ex CNG 105, lot 584”) join to 13983 earlier sales in the archive and give held-out same-coin pairs that are never used by the pipeline; on them the combined rule catches 0.797 of resales at a 0.6% false-flag rate, and the residual is carried with that recall’s uncertainty. Collapsing resales leaves 85058 specimens.

The verifier is calibrated in domain. 328 of Schaefer’s own coins occur in sixbid and carry his die labels: our pipeline accepts 59 of their 64 same-die pairs over all pairs (0.922) and 4 of 847 different-die candidate pairs. Recall on Schaefer’s catalogue scans is 0.47, so a bench-only calibration would have halved every count; photograph quality, not the method, sets recall.

The model is the one population genetics uses for effective population size. A type’s D dies each strike $c_i \sim \text{Gamma}(\alpha, \frac{\mu}{\alpha})$ coins; two coins share a die with probability $q = \sum p_i^2 \approx \frac{1+\frac{1}{\alpha}}{D}$, so the pair-share rate measures the *effective* die count $D_{\text{eff}} = \frac{D}{1+\frac{1}{\alpha}}$ and output is $T = (1 + \frac{1}{\alpha}) \frac{\mu}{q}$. Schaefer’s per-die counts fit a negative binomial with $\alpha = 2.68$ (Figure 4 C), so $D = 1.37D_{\text{eff}}$. The accepted-pair count is binomial in q , the recall and the false-positive rate; coins per obverse die is a log-normal prior spanning 10000–30000 (the numismatic convention); each type’s weight is bootstrapped from the upper quartile of its own lots’ published weights, fineness is 0.96 ± 0.015 , and each type’s production year is uniform over its catalogue date range. On the 9 Schaefer types present in the corpus the estimate runs at a median 0.862 of Schaefer’s count, with a type-to-type spread of about 0.7–1.2 (Crawford’s sub-letters are kept in the labels so that his sub-types can be compared on the lots that cite them). Since the pair decisions agree with his on the same coins, what remains is a difference between the two samples of each type, left open here and carried as a symmetric systematic (log-sd 0.149) that widens the interval rather than moving the estimate. All are sampled jointly in 4000 draws.

Figure 4 gives the posterior for 492 silver denarius types with at least 20 specimens: 2722 tonnes of silver struck between 211 and 31 BC (95%: 1427–5292 50%: 2181–3436), peaking at 52 tonnes in the year 85 BC. The two surges, the Social War and the civil wars of the 40s, are the ones the literature expects. The variance decomposition (Figure 4 B) is the point: 70.9% of the posterior log-variance is the coins-per-die prior and 21.6% the residual disagreement with Schaefer, while pair sampling, calibration, resales, weights and dating together contribute little.

The absolute tonnage therefore inherits a convention, and the argument about it (Buttrey’s objection that coins-per-die is unmeasured) is a fair one. The relative history does not. Every term that dominates the absolute number is a global multiplier, applied to every type alike: coins per die, the residual against Schaefer, the dispersion factor $1 + 1/\alpha$, fineness. In a ratio between periods they cancel exactly, and the assumption that replaces “we know coins per die” is the weaker “coins per die did not change systematically across a single denomination struck by one mint in one technique”. On that footing the results are sharp: the decade of the Social War (89–80 BC) struck 8.4 times the silver of the 140s (95%: 7.9–9.1), the civil-war decade of the 40s 7.5 times (7–8.1), and 44.1% of all Republican denarius silver (42–46.2%) was struck in the two war periods, 91–80 and 49–31 BC, thirty of the coinage’s hundred and eighty years. The die count is measured; what is left uncertain is one convention that the relative history does not need.

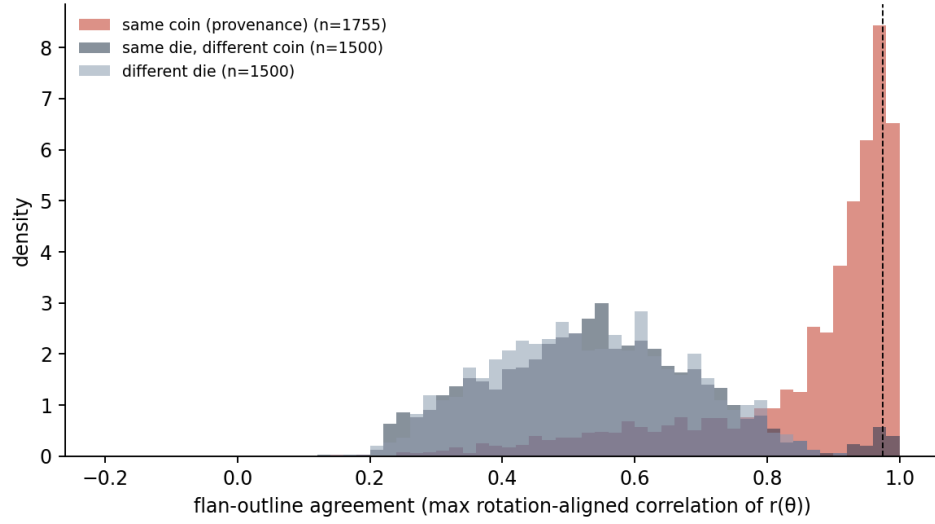


Figure 3: Same coin versus same die from the flan outline. Rotation-aligned correlation of the polar radius profile of the flan, for held-out provenance-linked resales (same coin), weight-different high-relief-score pairs (same die, different coin) and low-score pairs (different die). The outline carries no die information and identifies the coin; the small same-die tail near 1 is undetected resales in that reference set.

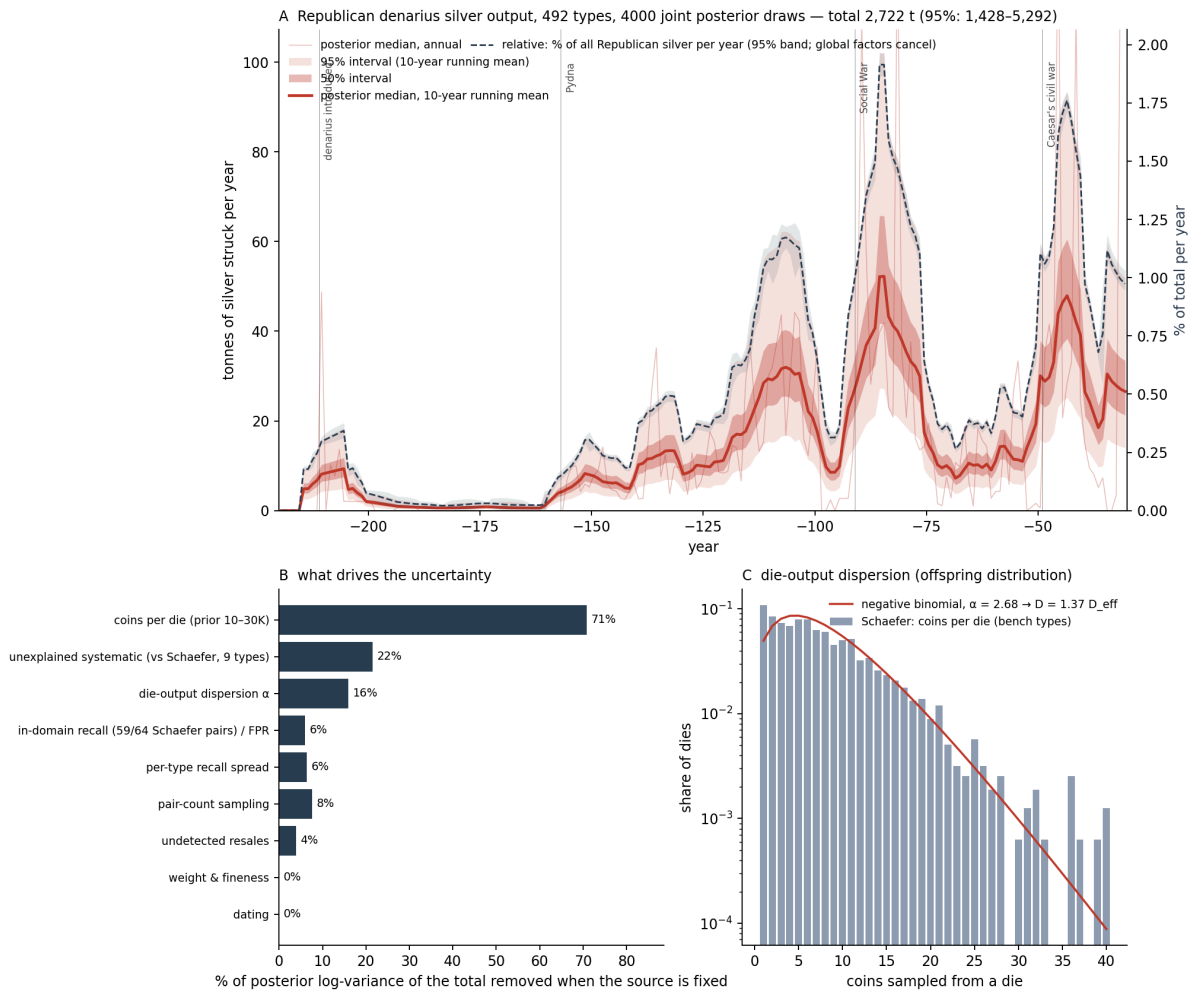


Figure 4: Republican silver output from die matches. (A) Tonnes of silver struck per year, 10-year running mean applied to every draw (annual median faint): posterior median with 50% and 95% intervals (red, left axis); dashed, the same draws normalised to their total (right axis, 95% band in grey), in which every global factor cancels. Posterior median with 50% and 95% intervals from 4000 joint draws over every source of uncertainty. (B) Share of the posterior log-variance of the total removed when each source is fixed at its centre. (C) Schaefer’s coins-per-die distribution on the bench types and the negative-binomial fit giving the die-output dispersion α , hence $D = 1.37 D_{\text{eff}}$.

5.6. The Denarius Across Five Centuries

The Imperial denarius can be measured the same way, and because it is the same denomination the two series join. 1166 RIC denarius types with at least 20 specimens were verified the same way (407098 lots, 392263 specimens after resale collapse). Two differences from the Republican case are worth stating plainly. First, Schaefer’s labelled corpus is Republican, so the Imperial estimate has no die ground truth of its own: the Republican systematic (log-sd 0.15) is carried as an assumption rather than measured, and the validation panel reports zero types. Second, Imperial dies span catalogue numbers where Republican dies do not. Sampling pairs that cross RIC numbers within one reign and mint, the acceptance rate relative to within-number pairs has a median of 0.05 across 32 series but reaches 1.18 for Tiberius, 0.56 for Galba, 0.29 for Otho and 0.21 for Vitellius, against 0.00 for Marcus Aurelius, Caracalla and Geta. RIC’s numbering splits die-linked issues for the Julio-Claudians and the civil war of 69, so pooling by number would inflate their die counts; the estimate divides by the measured factor per series.

Figure 5 shows the result. Output peaks at 13.7 million denarii a year in 85 BC, during the Social War, and the largest Imperial peak — 12 million a year under Vespasian in AD 73 — is 0.87 of it. The Republic at war struck denarii faster than the Empire ever did. Silver tonnage is drawn only where fineness has

been measured, so the series stops at Hadrian: the blank after AD 138 is the end of the evidence, not a collapse in output.

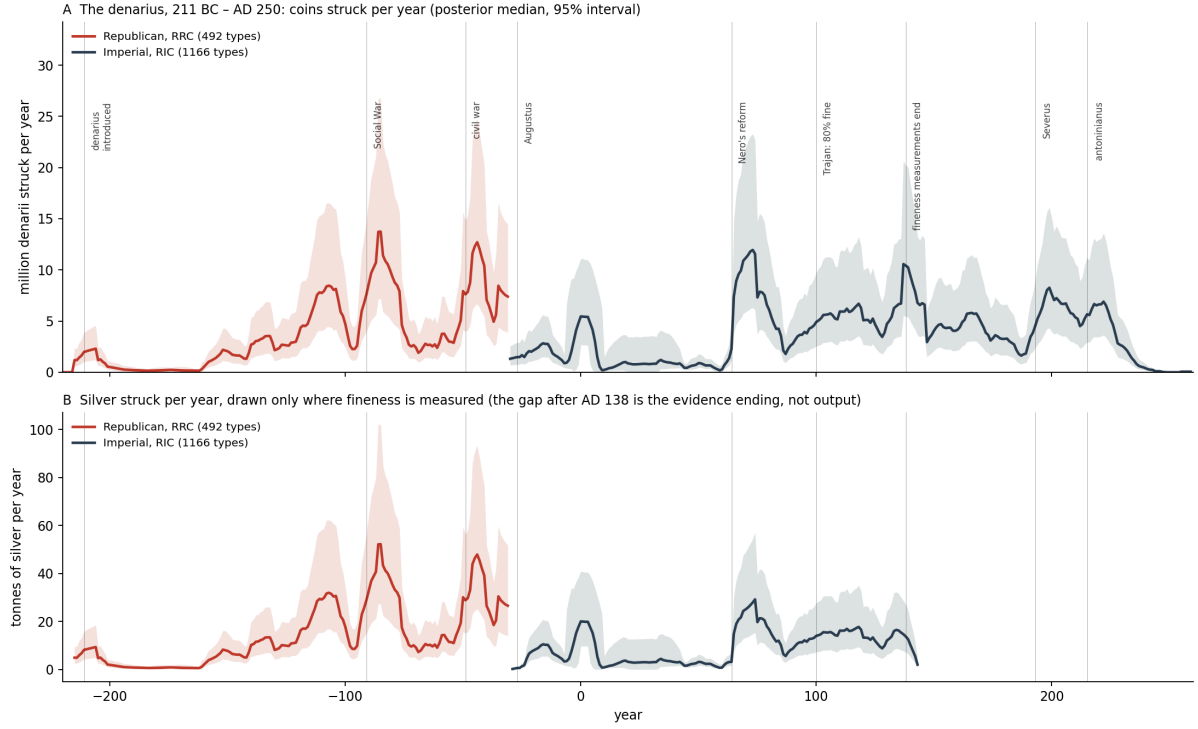


Figure 5: The denarius, 211 BC – AD 250. (A) Millions of coins struck per year, posterior median and 95% interval, for the Republican (RRC) and Imperial (RIC) denarius; the two share a method, a verifier calibration and a coins-per-die prior and differ only in catalogue. (B) Tonnes of silver per year, drawn only over the period for which fineness has been measured.

6. Experiments

6.1. Validation Levels

7. Results

On the frozen RRDP test split, geometric verification fused with the DINOv3-LoRA embedding separates die-mates from hard (same-type) negatives with verification AUC 0.9591 (obverse) and 0.9476 (reverse). Pairwise quality is high; converting trustworthy links into the correct die *count* — robust to the percolation cliff — remains the binding constraint (see Discussion).

8. Discussion

9. Conclusion

Bibliography